

Importance of basic science and research training for the future generation of cardiologists

A statement of the ESC Working Group on Cellular Biology of the Heart, the Heart Failure Association of the ESC, the ESC Council on Basic Cardiovascular Science, the European Association of Percutaneous Coronary Interventions of the ESC, the ESC Working Group on Cardiovascular Regenerative and Reparative Medicine, the ESC Working Group on Cardiovascular Surgery, the ESC Young Community and the ESC Scientists of Tomorrow

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Online publish-ahead-of-print 25 September 2025

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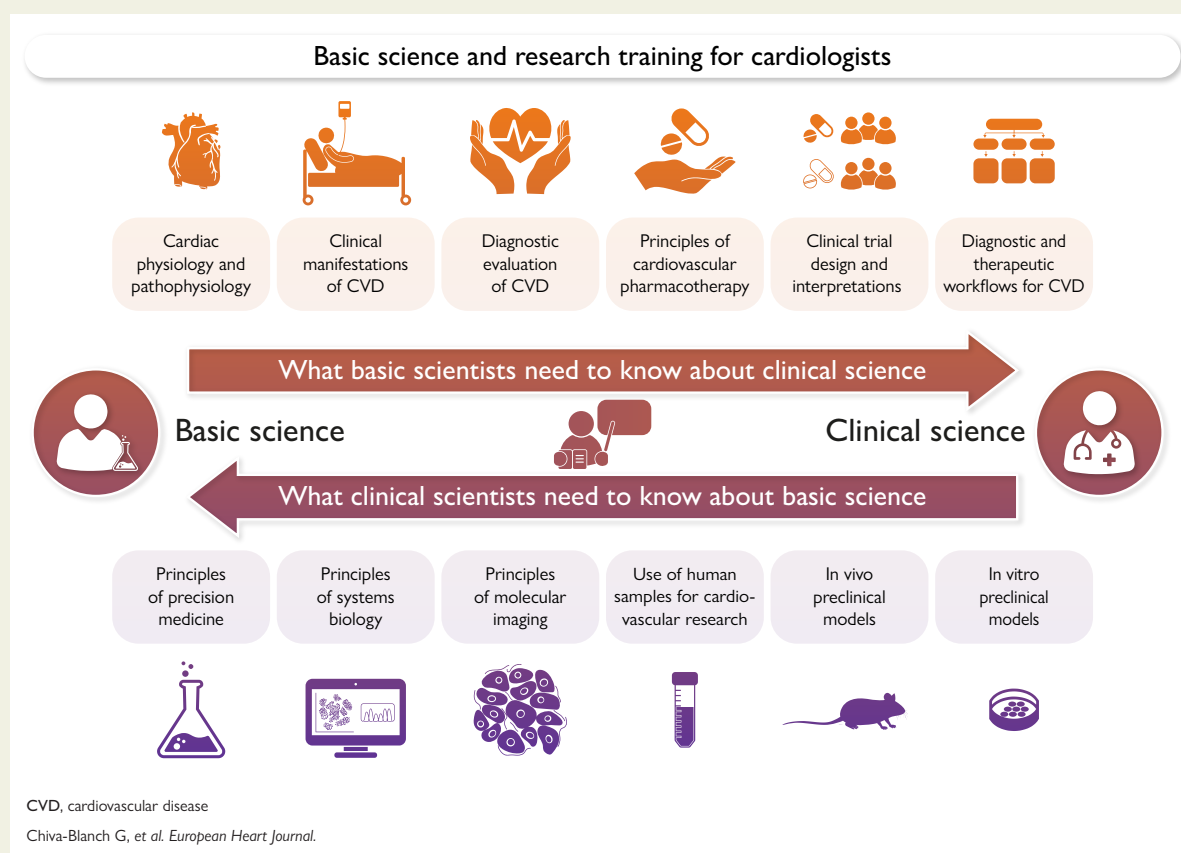
† This author suddenly passed away in January.

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Abstract

Advances in the battle against cardiovascular diseases depend upon continuously translating emerging scientific knowledge from preclinical studies and clinical trials into innovative and effective therapeutic strategies. Over the past three decades, molecular and cellular biology have undergone a profound transformation, and large-scale, single-cell, and multi-omics studies have enabled investigations on cardiac disease mechanisms with unprecedented precision. However, these rapid advancements have also contributed to a divergence between the needs and aspirations of basic researchers and those of clinical scientists and practitioners, to the detriment of discovery science, precision medicine, and cardiovascular healthcare. The present document highlights the importance of education and training in overcoming the gap between discovery and clinical science, by promoting a common language aimed at designing more translationally relevant and impactful discovery science. To achieve this aim, multidisciplinary efforts will be required to better define learning objectives within training programmes, including education in discovery and clinical sciences, promotion of specific mentorship paths, collaborative research efforts, promotion of equality, diversity, and inclusion, and protection of research time and activity within academic, clinical, and research careers.

Graphical Abstract



Bridging the knowledge gap between basic science and clinical cardiology. Education is crucial to bridge the widening gap between discovery science and clinical science. To reach this aim, basic scientists will need to strengthen their knowledge on cardiac physiology and pathophysiology, clinical manifestations of cardiovascular diseases (CVDs), clinical evaluation of CVD, principles of pharmacotherapy, clinical trial design and interpretations, and diagnostic and therapeutic workflows for CVD. Clinician scientists will also need to strengthen their knowledge on *in vitro* and *in vivo* preclinical models, types and conditions of human samples for cardiovascular research, principles of molecular imaging, systems biology, and precision medicine.

Keywords

Clinical research • Education • Cardiology • Translational research • Basic research • Cardiology training

Introduction

Given the rapid and constant evolution of medical knowledge, science and research must serve as foundational pillars for general and specialty education and training.¹ Regrettably, the number of physicians who

choose to dedicate (at least in part) their career to the pursuit of scientific endeavour is declining worldwide.^{2–4} Consequently, the future of academic medicine has become a topic of intense debate.⁵ The rapid pace of development in basic science over the past 30 years has paradoxically contributed to the progressive divergence between expertise

and careers in basic research, clinical research, and practice.⁶ Although basic science remains a fundamental component of most medical school curricula, rigorous scientific education is increasingly omitted from cardiology fellowship programmes in favour of clinical training. Consequently, it is unsurprising that a decline in the number of basic science studies published in medical journals has been observed in recent years.^{7,8} The gradual dissipation of basic science reports from the medical literature and the disappearance of fundamental science from continuing medical education programmes, progressively erode the mastery of science within the medical profession.⁹ Limited translational perspectives and education in clinical sciences during master's degrees or PhD in discovery sciences also contribute to the widening gap between basic, translational, and clinical research.

Integrating expertise is essential to bridge the gap between evolving scientific knowledge and its translation into effective clinical applications. Education and training in basic, translational, and clinical sciences are key to this process.^{10,11} Although 'education' and 'training' are sometimes used interchangeably, they hold distinct meanings, costs, and practical implications concerning professional development. Education is a long-term, continuous process, whereas training is usually more focused, shorter in duration, and provides competence in specific, more practical skills. While most practicing clinicians will not require practical training in the techniques of basic and translational science, they might still need to acquire or update general theoretical knowledge on mechanisms of disease, the action of drugs, types and conditions of human samples for cardiovascular research, principles of molecular imaging, systems biology, and precision medicine. Without this knowledge, they will have a superficial understanding of scientific publications' significant results and translational implications, and they will find integrating basic, translational, and clinical information harder to achieve optimal clinical outcomes. In fact, basic scientists in cardiology should also acquire or strengthen their knowledge on cardiac physiology and pathophysiology, cardiovascular disease (CVD) manifestations and therapeutics, clinical trial design and interpretations to effectively share this common language and design more clinically, translationally relevant, and impactful discovery science.

One of the aims of the European Society of Cardiology is to promote programmes in support of continuing medical training and education in the rapidly changing scenario of cardiovascular medicine. The purpose of this document is to provide multidisciplinary insights and suggest actions to integrate discovery, translational, clinical science, and clinical practice through education. Although only a proportion of all clinicians will become clinician scientists, facilitating the interaction between these areas of expertise in the educational and training pathways is expected to foster collaboration and scientific curiosity, promoting a common language aimed at designing more translationally relevant and impactful research.

The future cardiologist

To be effective physicians, the future generations of Cardiologists will need to understand the translational implications of scientific publications in high-impact and broad interest scientific journals. Based on the chosen cardiovascular sub-specialty, knowledge of basic science methods and techniques, appropriate preclinical models, and the interdependence of the available devices with cellular functions and electrophysiology, might also be required. Clinical/translational research also increasingly relies on Big Data acquisitions and analysis. Therefore, competencies such as coding, programming, artificial intelligence (AI),

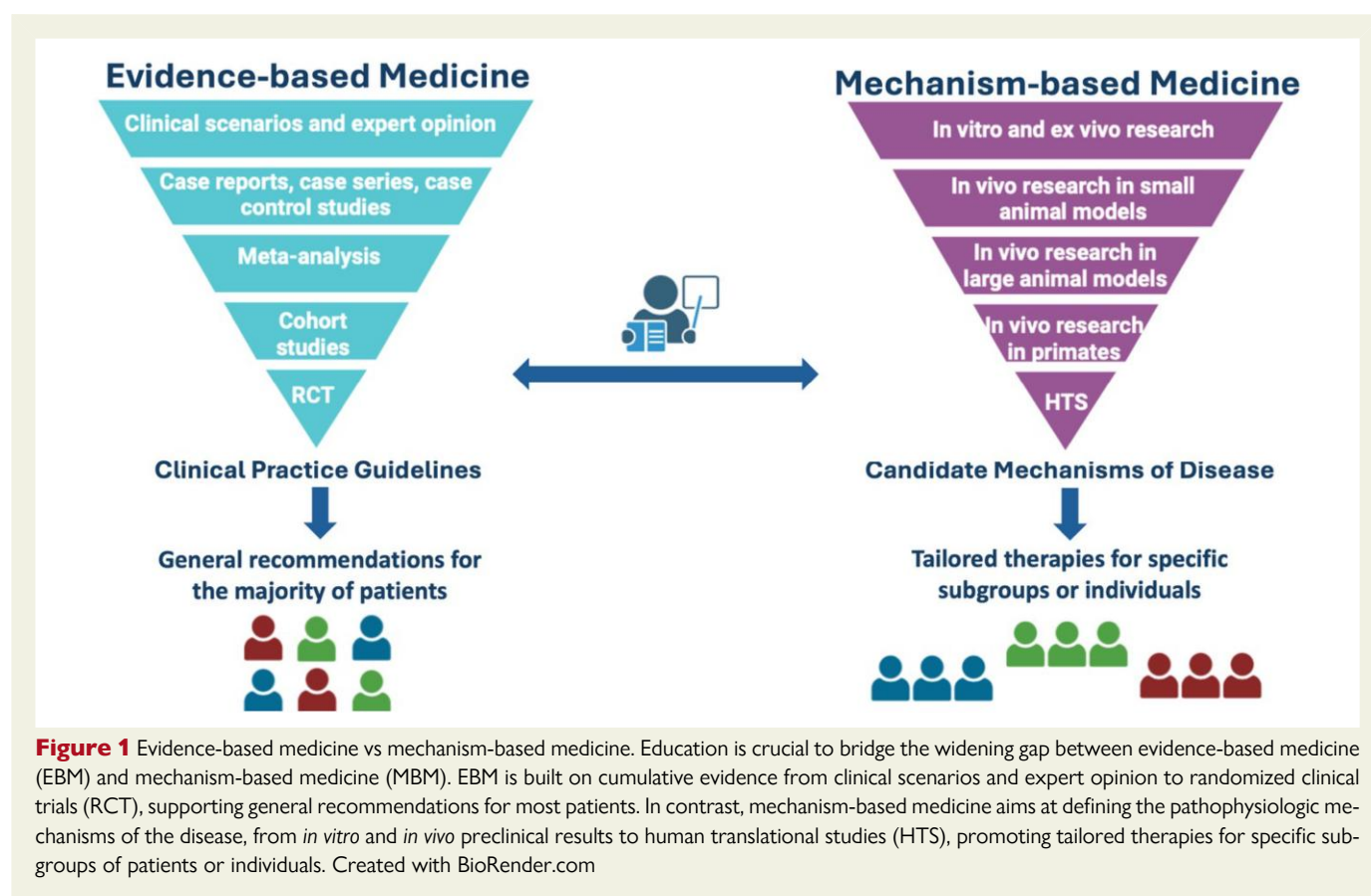
bioinformatics, and machine learning are becoming mainstream activities in specific research paths. In fact, one of the emerging new skills that the future cardiologist might need to acquire is the use of AI and the critical evaluation of its outcomes. AI is changing the way both research and healthcare is conducted, not simply as a tool, but also as a revolutionary force by identifying novel therapeutic targets, predictive modelling, and efficient data integration, as considered in the AICare@EU (Deployment of AI in Healthcare), in which prevention, diagnosis, treatment, and rehabilitation of cardiovascular diseases is listed as a priority area.¹² However, to guarantee the successful and efficient deployment of AI tools and to increase their equitable and fair adoption in research and clinical practice, a number of obstacles still need to be overcome, such as ethical and legal aspects or the intrinsic bias of some AI algorithms. Achieving this level of knowledge will require alignment between clinical practice, research, and teaching interests during the educational period to ensure a comprehensive integration of concepts.¹²

Early engagement in research during medical school or cardiology training is crucial, as timing within education can significantly shape learning and career trajectories. Early research involvement builds skills in study design, data analysis, and scientific communication, and also represents an opportunity to explore cardiology sub-specialties—such as electrophysiology, heart failure, or interventional cardiology—helping trainees identify specific interests and align their research accordingly in subsequent years. This focus sharpens expertise, guides fellowship choices, and positions trainees for future leadership.

Importance and complementarity of basic and clinical sciences in Cardiology

Evidence-based Medicine (EBM) approaches are now widely employed in current cardiology education and training programmes, utilizing clinical practice guidelines developed through consensus based on randomized clinical trials conducted in selected patient populations under specific and highly regulated conditions (Figure 1, left panel).¹³ Despite the recognized and unrefusable value of EBM, translating the results of clinical trials into the real world of specific patients is often challenging. Additionally, EBM does not necessarily test or elucidate the mechanism(s) underlying the observed biological or clinical effects. For example, inhibitors of the Sodium-Glucose Cotransporter-2 (SGLT2i; gliflozins) were designed to promote urinary glucose excretion by inhibiting glucose re-uptake in the kidney proximal tubule, and were originally prescribed as oral antidiabetic drugs, yet subsequently demonstrated impressive beneficial cardiovascular effects in randomized controlled clinical trials,^{14–17} even though the precise mechanisms responsible for these effects remain incompletely understood.

The aim of basic research, often referred to as 'pure', 'fundamental' or 'discovery' research, is to investigate the mechanisms of biology, physiology, and disease. It is driven by curiosity or interest in scientific questions, and it aims at expanding knowledge without necessarily creating or inventing new technologies or treatments. By enhancing the understanding of living systems, basic research ultimately contributes knowledge that can be employed to improve predictions, to prevent, diagnose, and treat diseases by identifying candidate mechanisms and tailored therapies for specific subgroups

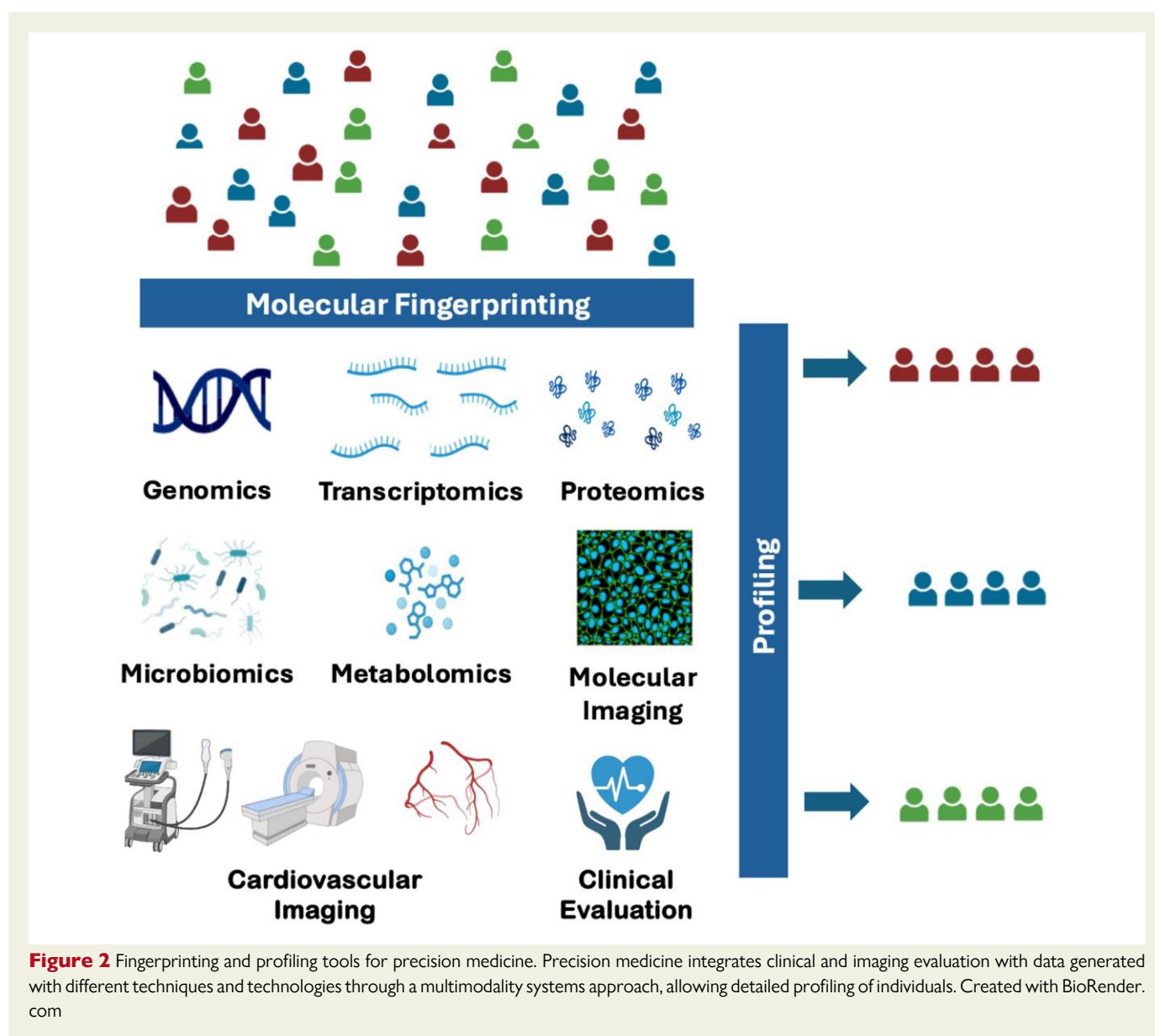


or individuals (Figure 1, right panel). However, the time lag between biomedical research findings and their translation into health improvements has been variable and it is usually estimated in decades, including the final clinical testing in trials.^{18,19} Therefore, basic science can be perceived as distant from the patient and is progressively met with declining interest among clinicians.⁷ Nevertheless, several historical examples provide strong evidence for the fundamental role that basic science had in many diagnostic, prognostic and therapeutic innovations in cardiology.^{20–22} Classic advancements in research include genetic testing, biomarker discovery and imaging techniques, all of which have significantly contributed to the understanding of the aetiology of CVD. Currently, exciting advances are also being made in molecular imaging, with new probes that can image biological processes in the whole animal or the patient.²³ New developments in single cell sequencing and multi-omics (genomics, epigenomics, transcriptomics, proteomics, metabolomics, and microbiomics) for deep phenotyping, have provided valuable insights towards precision medicine (Figure 2).^{24–26} The relationship between the patient, the disease, the treatments, and the environment can significantly influence health status and recovery from illness.²⁷ Additionally, incorporating AI and machine learning techniques is essential,²⁸ especially when considering the socioeconomic inequalities that may affect their development and effectiveness.²⁹ By acquiring and integrating data from individuals, precision medicine aims at shifting from the current one-size-fits-all medical model (EBM alone), to a treatment paradigm tailored to each individual or group of individuals with similar pathophysiological mechanisms. In this new era of precision medicine, specific educational programmes will be necessary to provide an integrated translational approach.

In complex clinical scenarios, mechanistic reasoning can enhance decision-making capabilities and interpretations of a broad range of symptoms seen in patients, especially in complex or unexpected situations.³⁰ Indeed, acquiring mechanism-based, systems biology approaches, as well as clinical evidence-based knowledge can enrich clinical training and future experience, shaping different research possibilities (and career paths) during physician's rotations and beyond.

Another advantage of clinicians having a mechanistic understanding of basic and translational science is that this will provide them with the ability to recognize the significance of unexpected clinical observations. This recognition can facilitate the 'bedside to bench' or 'reverse translation' of clinical results. Such an approach can inspire initiatives that lead to the direct repurposing of therapeutic tools, thereby shortening the time between basic observations and clinical application.¹⁹ Establishing a common language among professionals in basic, translational, and clinical research enhances communication and understanding between scientists and medical practitioners, or even across different fields, facilitating the translation of laboratory discoveries into clinical applications for the benefit of the patients.

Finally, education in basic sciences and practicing mechanism-based research teaches to assess studies and observations critically, promotes independent thinking, accuracy, honesty in interpreting data, and, most of all, scientific curiosity. In addition to intelligence and effort, intellectual curiosity has been proposed as the third pillar of academic achievement, positively predicting future academic performance.³⁰ Heightened curiosity in medical students is positively linked to memory, the ability to identify mistakes, solve problems and is a precursor to empathy.³¹ As previously stated, not all clinicians will become clinician scientists. In this



context, identifying and nurturing such individuals from an early stage in medical training is crucial to increase the proportion of clinical scientists and optimize their training.

Education in medicine and cardiology in Europe

Cardiology is a highly specialized field within internal medicine, and the curriculum for the specialization in cardiology generally encompasses a combination of theoretical coursework, practical training, and clinical rotations. The duration of the programmes can range from around 4 to 10 years, depending on the country and the specific programme (Figure 3). Studying medicine or Cardiology in Europe can provide students with a multicultural and diverse learning environment, allowing them to gain a global perspective and develop cross-cultural skills. However, given the disparities in the different European countries^{32,33} (see [Supplementary data online, Table S1](#)) and available education pathways (Figure 3), it is difficult to ensure equivalency in the standards for the same title (MD, Cardiologist, PhD or MD/PhD) across countries.

The ESC core curriculum for the Cardiologist³⁴ states that 'Trainees must acquire the skills to critically evaluate new research and its implications for clinical practice. They should also develop a detailed understanding of the design and conduct of basic, translational, clinical, and epidemiological research either in their training institution or in collaboration with other centres or universities'. However, this curriculum primarily outlines the clinical competencies expected of cardiologists. Consequently, there is an unmet need for a genuine integration of research and basic science within the core curriculum for cardiologists, as well as a clear definition of the minimal competencies in basic and translational science that future cardiologists should acquire during their education and training.

Existing European initiatives to promote integrative education and training

A position paper of the Academy of Medical Science in the UK in 2007 recommended the integration of PhD programmes as a valuable option for training clinical scientists, encouraging the governmental

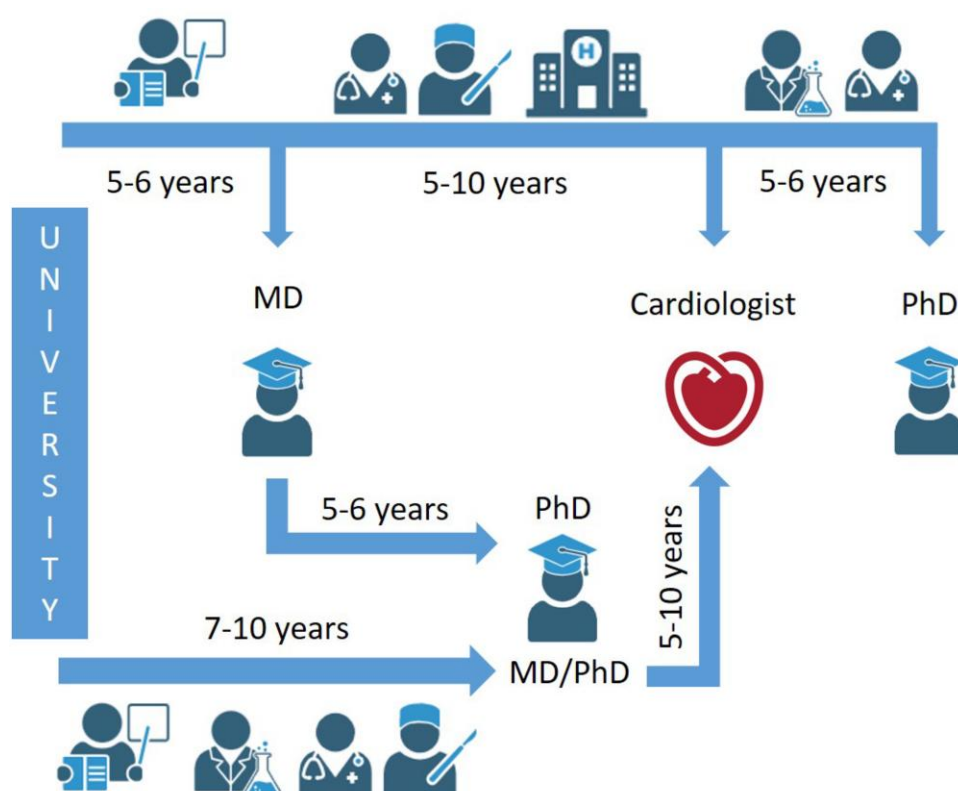


Figure 3 Possible education pathways in different European countries. Depending on the ESC country of study, there are different pathways to obtain the Medical Doctor (MD) degree, the Cardiology speciality and/or the Philosophical degree (PhD). There are significant disparities between countries in the minimum time to obtain each degree. Created with BioRender.com

programmes and industry for mentoring and support, pointing out the importance of exposing all medical students to scientific research.³⁵ Since then, much progress has been made in the implementation of all kinds of support for young scientists (i.e. paid clinical-free periods for research, empowering programmes for women in science, and child-care support for young scientists' families). However, the professional associations in Europe still point out the necessity of protected research time during the specialistic training for interested young medical doctors, to increase family-friendly career options for these researchers and to establish the legal environment in Europe for dual leadership (as existed in the USA)—i.e. separate managerial functions responsible for care and research.³⁶ So far, these programmes are implemented only in a few countries, and even only in some universities, varying across countries in terms of curriculum, duration, entry requirements, and accreditation.³⁷ Some examples of successful integrative programmes at different educational and training levels are provided in [Supplementary data online, Table S2](#).

In several European countries, combined MD/PhD programmes have been established in which students obtain both titles simultaneously. MD/PhD programmes appeared in Europe in 1989, first in the UK, and spread slowly around Europe (<https://www.eumdpd.com/mapping-european-md-phd-programs/>). The combination of MD and PhD training allows cardiologists to approach patient care holistically. The primary concern regarding implementing of such programmes in European universities is the belief that clinical skills may stagnate or be lost during time away from clinical practice. As a result, many educational systems permit attendance in these programmes but do not

recognize them as part of clinical training. Consequently, they may be less appealing for young clinicians to pursue during their clinical training. Although lengthening their study time, these programmes also allow them to approach basic research directly. PhD duration varies from 2 to 8 years depending on the country and the type of activity engaged by the student. It might largely differ from basic or translational PhDs in terms of duration, hands-on time invested in the bench and interpretation of results.

After completion of their education and training, a key challenge for clinicians and physician-scientists is the lack of time to conduct research alongside their clinical duties. For this reason, many medical faculties at public universities in Europe have introduced funding programmes for clinician scientists,³⁸ offering release from clinical duties and a curriculum for interdisciplinary training, including modern clinical and basic investigational methods and mentoring.³⁹ These programmes are, however, not a widespread standard across European countries.

Strategies to raise awareness, develop and support specialized career paths including research education

In this section, we highlight critical issues in this context and propose different strategies at several levels that scientific organizations, educational institutions, and policymakers can adopt. [Table 1](#) outlines

actionable initiatives to bridge the gap between discovery research, clinical research, and clinical practice across three domains: governance, education, and individual roles. In governance, it emphasizes fostering collaboration through shared research hubs, mentorship programmes, and aligned incentives, while supporting interdisciplinary teams and data sharing. Educational strategies include integrating clinical and basic science training, promoting joint seminars, and developing interdisciplinary degree programmes and certifications. On an individual level, scientists and clinicians are encouraged to engage in cross-disciplinary learning, collaborative research design, and participation in translational initiatives to align research with clinical needs. We expect that a combination of these strategies and initiatives might build a robust ecosystem that nurtures and sustains research career paths, ensuring continued innovation and excellence in research, education, and clinical practice.

Policy strategies

In strategically mapping an EU core curriculum integrating basic and clinical science in medicine and cardiology, the need for essential policies becomes evident.^{40–42} Establishing career paths that seamlessly integrate clinical and research components, with a strong emphasis on a solid foundation in basic science, including statistics, is essential for advancing translational progress in cardiovascular medicine. However, to date no concrete activities have been materialized.

Core curricula and data sharing

To address the significant differences in research education and training across various cardiology programmes in Europe, one potential strategy is to develop a standardized core curriculum. This curriculum should provide guidance on this essential aspect of training and include input from all 27 European countries. However, the fact that some European countries still do not have an internal (within the country) core curriculum programme hampers this idea. Correspondingly, the efforts from National Cardiac Societies for a common programme to enable homogenized training are crucial. Merging the ESC Core Curriculum³⁴ in the frame of the European Higher Education Area would also promote a better exchange of educational programmes between institutions and countries. In the meantime, Universities might offer micro-credentials⁴³ to give the opportunity to MD and cardiologists to acquire the minimum competencies to switch to a mixed approach in their daily working activities.

Certification of education and training

Linking education and training with official acknowledgement of specialization through certification would increase eligibility for support, employability, and career advancement. However, the implementation of such certification in EU is currently a challenge.^{44–46} For clinicians, certification of participation in specific basic science CME courses is in some countries expected and registered; this is not the case for translational scientists. A common cardiovascular curriculum for clinical scientists, jointly developed by the ESC and national cardiac societies, would be a major step towards training translational scientists and their field needs.

The ESC adeptly crafts comprehensive career pathways that seamlessly integrate clinical and research aspects, recognizing the pivotal role of fundamental science in driving translational breakthroughs in cardiovascular medicine. The ESC's 2024–28 strategy document supports it as a guiding force steering cardiovascular education towards innovation and scientific rigour. This aligns seamlessly with the missions of esteemed organizations such as the National Institute of Health

(NIH),^{44,46} the American College of Cardiology (ACC),⁴⁵ and the American Heart Association (AHA). Collectively, these entities epitomize a shared commitment to nurturing the forthcoming cohort of physicians. At the core of these shared strategies lies an acknowledgment of the indispensable role of basic science in moulding the training of future cardiologists, which is pivotal for the advancement of translational breakthroughs in cardiovascular medicine. Consistently, there is a clear and strong commitment of the current and upcoming ESC leadership to consolidate and further promote international educational programmes in basic sciences. For example, the biennial ESC CBCS Summer School (<https://esc365.escardio.org/event/1867>) brings together early-career basic and clinical scientists to educate them on vascular and cardiac science and disease. It also offers valuable mentoring sessions. The main objective of ESC CBCS Summer School is to inspire a new generation of cardiovascular science leaders among highly motivated science and medical graduates. Expert faculty featuring prominent and upcoming leaders in their fields give educational presentations, mixed with workshops and mentoring sessions, as well as poster presentations and pitches by the young researchers. This highly attractive format is familiar, dynamic, and interactive, and provides a welcoming atmosphere with time for poster presentations, scientific discussions, and informal social interactions. It is a great networking opportunity for both basic and clinical scientists to learn from each other and foster new networks for future collaboration. Expanding these initiatives to reach a larger number of early-career researchers would be a highly beneficial step.

Educational and research funding

Funding schemes for academic programmes that integrate basic science with medical or cardiology training vary depending on the country, institution, and specific area of focus. Nonetheless, several funding sources exist to support such interdisciplinary training programmes. Some universities and research institutions offer internal funding schemes with scholarships, fellowships, and research grants for interdisciplinary training programmes. In addition to government grants and fellowships, non-governmental organizations and charities provide opportunities for cardiovascular fellowships and scholarships, but this is usually limited to residents of that country. For instance, the Erasmus Programme during the first years of the training period might represent an excellent opportunity to foster internationality and the first experience in basic science. Horizon Europe is the EU's key funding programme for research and innovation. The Marie Skłodowska-Curie Actions (MSCA) are the European Union's reference programmes for doctoral education and postdoctoral training. The MSCA aim to support researchers at all stages of their careers across all disciplines, including the medical and health sciences. The ESC Council on Basic Cardiovascular Science (CBCS) of the ESC offers grants and a fellowship programme. These include the ESC First Contact Initiative Grant to promote careers in basic cardiovascular sciences in Europe and the ESC Basic Research Fellowship. However, the number of these grants available each year is extremely limited and exclusively for the postdoctoral period.

Overall, funding of fundamental 'basic' research has declined compared to translational or applied research in many countries.⁴⁷ This shift could result in a rise in translational or applied projects that tend to be more derivative of existing knowledge, potentially lacking originality and ground-breaking insights. In fact, only 6%–8% of the total submitted abstracts to the annual ESC Congress in the last 10 years are focused on basic science, with remarkable geographic variations (Figure 4A). In

Table 1 Examples of actionable initiatives to bridge the gap between discovery research, clinical research, and clinical practice

Governance			
Short-term		Mid-term	Long-term
- Implement programmes where scientists spend time in clinical settings to observe and understand real-world clinical needs and workflows, and clinicians can be involved in research activities or collaborations. - Create hubs where clinicians and basic scientists work together on disease-focused projects. - Preserve and protect research time and activity within clinical career paths. - Provide technical and scientific support for research activities (Grant Office, Patent Office, etc.). - Promote regular meetings between scientists and clinicians to discuss ongoing clinical challenges and research developments, to foster mutual understanding and collaboration. - Promote initiation of pilot programmes (internal grants) for joint research projects between clinicians and scientists. - Establish incentives and awards to recognize outstanding collaborative research efforts and contributions to the field.		- Promote specific mentorship programmes, educational tools and career tracks. - Monitor and support joint research projects that involve both scientists and clinicians, ensuring that clinical insights are integrated into research design and implementation. - Facilitate mutual learning and networking across specialties. - Promote equality, diversity and inclusion in career promotion and retention. - Create aligned incentives and awards for research activity and productivity in tenure-track positions. - Foster research teams that include both basic scientists and clinicians. - Enable and support shared roles across multidisciplinary departments. - Provide institutional support for data sharing, by building secure platforms for sharing clinical data with basic researchers while ensuring compliance. - Establish career bridging programmes (for example 'Translational Scholar' tracks) with seed funding for collaborative projects).	- Secure 'high-risk' and highly innovative research. - Increase research funding opportunities, consider corrections or incentives based on sex and gender, specific familial conditions, combined research and clinical practice activity. - Reward interdisciplinary publications, patents, and grants in promotion/tenure track criteria. - Work with EU policymakers to develop and implement collaborative research and innovation in cardiology at the EU level, including funding for interdisciplinary projects and infrastructure that facilitates collaboration. - Ensure the sustainability of interdisciplinary initiatives and continuously evaluate their impact on clinical outcomes and discovery research advancements. - Establish translational research centres.
Education			
Short-term		Mid-term	Long-term
- Integrate basic and clinical science in scientific congresses. - Schedule regular joint seminars and journal clubs. - Crosstrain students and fellows. - Encourage clinical exposure in PhD programmes and basic science modules in medical residencies/fellowships. - Offer short-term lab or clinical rotations. - Organize cross-training workshops to align understanding of research questions, priorities, and terminology.		- Make science and research more available to students throughout the educational and training periods, allowing trainees to choose integrated education programmes combining clinical training or education with research skills. - Expand research and clinical training in single or combined M.D./Ph.D. degree programmes, and interdisciplinary postdoc/residency tracks. - Foster co-mentorship models, by assigning mentors from both clinical and basic sciences to trainees.	- Define national and international certifications to promote and facilitate international exchange. - Define research core curricula and career tracks. - Develop shared language and goals. - Establish joint PhD/MD programmes with integrated curricula with translational science modules.
Individual			
	Short-term	Mid-term	Long-term
Basic scientists	- Shadow clinicians (spend time in clinics, surgical suites, or inpatient rounds to understand patient care pathways). - Learn basic clinical concepts. - Take online courses or attend medical grand rounds to gain clinical context. - Read clinical case reports or guidelines to identify unmet needs your research could address.	- Attend interdisciplinary conferences and introduce your work to potential clinical collaborators. - Align research questions with clinical relevance. - Volunteer for hospital translational boards.	- Get involved in clinical trial design or ethics committees. - Co-author with clinician scientists and clinicians. - Co-design research with clinician scientists and clinicians.

Continued

Table 1 Continued

Individual			
	Short-term	Mid-term	Long-term
Clinical scientists	• Improve understanding of basic science language and tools. • Participate in local research seminars and lab meetings with mixed expertise. • Reach out to local researchers to explore collaboration.	• Search collaborations with a lab team or research group. • Gain some hands-on experience with basic science methods. • Push for dedicated clinician-researcher time. • Lead journal clubs or teaching sessions focused on bridging research and clinical practice.	• Co-author and co-design research with basic scientists. • Help establish research-integration units or translational centres. • Bring real-world patient challenges into the lab setting to guide meaningful research.
Clinicians	• Improve understanding of basic science language and tools. • Participate in interdisciplinary seminars or grand rounds with research themes. • Create formal pathways between fundamental research labs and clinical environments.	• Identify and refer eligible patients. • Promote awareness within your clinic. • Bring basic science results into clinical context and real-world patient challenges.	• Use mechanism-based reasoning to solve complex clinical questions. • Act as a Principal Investigator or site leader for studies translating discovery into practice. • Influence policy using outcomes data and translational metrics.

addition, there has been a progressive decline in basic science abstract submissions to the ESC Main Congress from ESC countries and Japan in the last four years, associated with a simultaneously increased rate of submissions from China (Figure 4B), while submissions of case reports have surged, indicating a disturbing shift away from basic science interest. This shift could also be contributed by the continued cultural acceleration of the pressure for researchers to ‘publish or perish’.⁴⁸ Typically, publishing a case report is a faster process compared with the publication of an original research manuscript.

To prevent this progressive decline in basic science research, securing high-risk and innovative research will be necessary by enhancing dedicated funding calls and selections. To support the development and advancement of research careers for physicians, it would be beneficial to establish rewards and recognition for research activity and productivity in tenure-track positions. Additionally, increasing opportunities for research funding and considering adjustments or rewards related to clinical duties and other factors could also contribute to this goal.

Sex and gender differences in career paths

Sex and gender are important variables that influence all aspects of cardiovascular clinical practice, education, and training,⁴⁹ including the involvement of women in cardiovascular research.^{50,51} Despite significant efforts in terms of gender equality, women remain underpaid,⁵² under-represented in science and medicine, especially in leading roles,⁵³ are less likely than their male peers to receive independent funding, produce publications, or retain academic positions.⁵⁴ Interestingly, women are increasingly likely to mentor women with future leading roles, increasing diversity and creating more inclusive research teams.⁵⁰ A set of recommendations aiming at ensuring that EU research will promote gender equity was elaborated by the ESC in collaboration with the European Parliament in 2018.⁵⁵ The ESC offers grants to attend the innovative Women Transforming Leadership Programme, an education programme focused on organizational challenges and identification of personal

leadership style. Moreover, the ESC Gender Policy was introduced in 2022 to correct the imbalance between the representation of women and men within the ESC and its constituent bodies. Also, in this line, the ESC Gender Task force was created in 2024 to mitigate gender inequalities among the different ESC structures but also to support women in cardiology and leadership roles and cardiovascular research in women. These initiatives have already greatly impacted for example the gender balance in active roles at the ESC Congress (in 2024 for the first time, women represented 50% of all faculty roles). Other achievements have been done, as it is the increase in female representatives among different ESC bodies as the ESC Board, ESC committees, ESC publishing teams. Even if these results are still not in the aimed goal, a clear positive trend can be noted. In fact, the long-term goal of the ESC Gender Policy is, besides promoting gender balance (40%–50%) in ESC structures, reducing gender bias both in tenure-track, research and clinical practice, through mentoring activities, educational programmes, acknowledgement of institutions with implemented gender policy, encouragement of journals to discuss about gender balance, overall promoting a cultural change towards gender balance in a transversal perspective. These ambitious objectives involve a profound structural change and will be further implemented in the short- and long-term.

Importance of academic mentoring

Academic mentoring is crucial for career development,⁵⁶ providing invaluable guidance,⁵⁷ models and support to individuals navigating the complexities of academic and research environments.⁵⁸ Mentors offer insights, advice, and encouragement, helping mentees navigate options and opportunities, evaluate challenges, make informed decisions, and stay motivated in their pursuits and goals. A good mentor should provide opportunities for skill development, offer constructive feedback, and facilitate networking and collaboration, all of which are essential for career growth and success in both the clinical and the research setting.⁵⁹ Academic mentoring often represents a unique opportunity for the mentee to test his/her

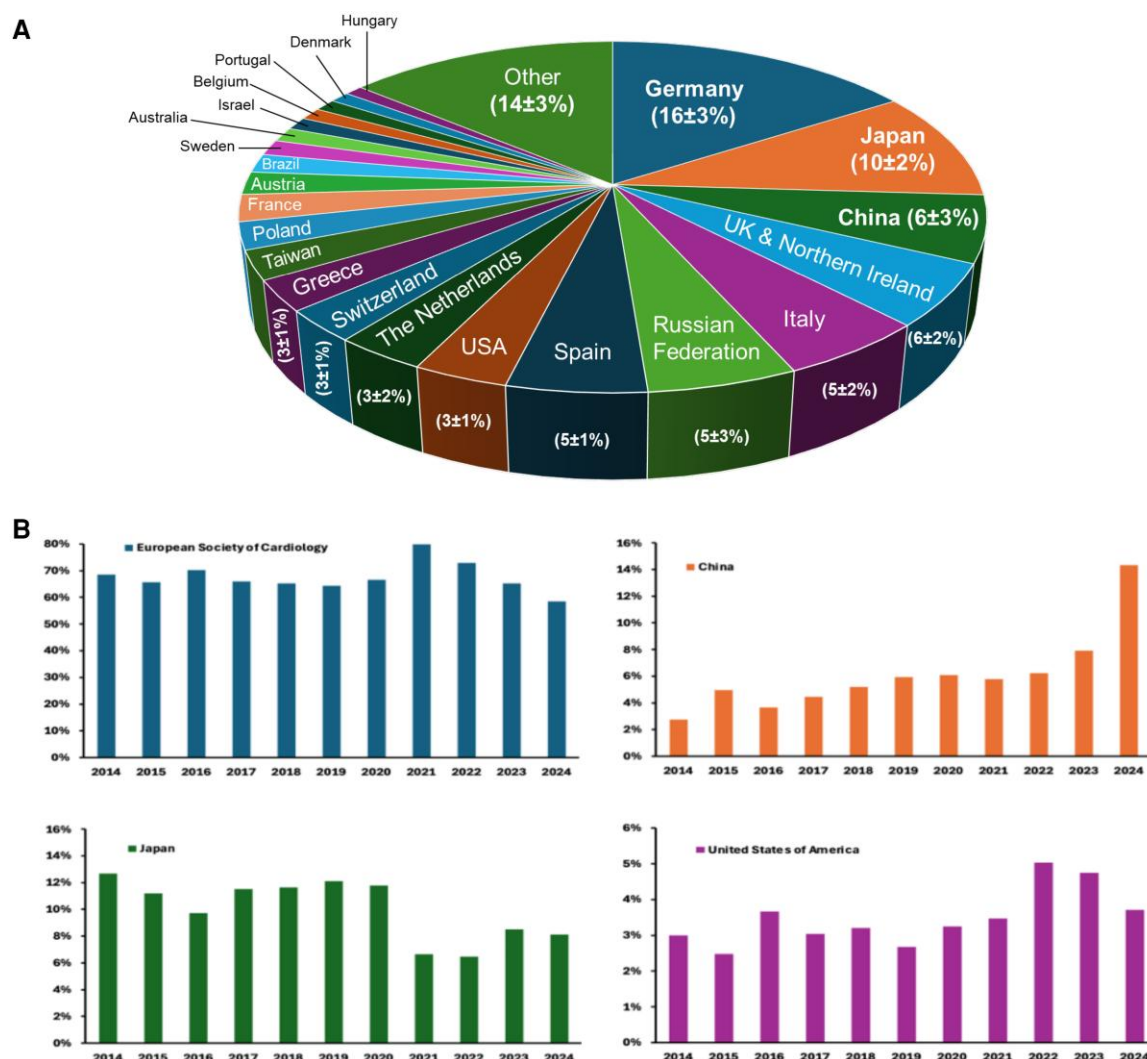


Figure 4 Geographic distribution of basic science abstract submissions to the annual ESC Congress (2014–2024). (A) Geographic distribution of major contributors of basic science abstract submissions to the Annual ESC Congress for years 2014–2024. Data shown are average $\pm$ standard deviation. (B) Percentage of basic science abstracts submitted from different regions accounting for 84%–96% of total basic science abstracts submitted annually. ESC countries include Albania, Algeria, Armenia, Austria, Azerbaijan, Belgium, Belarus, Bosnia and Herzegovina, Bulgaria, Croatia, Cyprus, Czechia, Denmark, Egypt, Estonia, Finland, France, Georgia, Germany, Greece, Hungary, Iceland, Ireland, Israel, Italy, Kazakhstan, Kosovo Republic of, Kyrgyzstan, Latvia, Lebanon, Libya, Lithuania, Luxembourg, Malta, Moldova (Republic of), Montenegro, Morocco, North Macedonia, Netherlands, Norway, Poland, Portugal, Romania, Russian Federation, San Marino, Serbia, Slovakia, Slovenia, Spain, Sweden, Switzerland, Syrian Arab Republic, Tunisia, Türkiye, Turkmenistan, Ukraine, United Kingdom of Great Britain (UK) and Northern Ireland, and Uzbekistan. Extracted from ESC archives

own attitude and curiosity towards research. Effective advising allows the student to gaze into the yet 'unknown' world of research, and is essential to inspire, create opportunities, and nurture the next generation of clinicians, researchers, and leaders in the field of cardiology. The ESC has recently started to offer a mentoring programme for its members supported by an easy-to-use online platform and will further promote this and other initiatives with a major focus on cardiovascular basic and translational research. In addition, several ESC journals also offer or are introducing peer reviewer and editor mentoring programmes, including European Heart Journal (EHJ), Cardiovascular Research, and EHJ Case Reports, which also provide useful tips and tricks both from the reviewer/editor and the author perspective.

Fostering curiosity during cardiovascular education and training

Curiosity drives individuals to seek out new information, skills, and perspectives. Several strategies can be undertaken to spark and leverage curiosity in students and trainees, mainly under mentoring guidance, creating time and opportunities to consider different scientific perspectives, making science and research more available throughout the educational and training periods, encouraging self-directed exploitation and learning.⁶⁰ Fostering curiosity also stimulates the capacity to generate new ideas and formulate new scientific questions when interacting with patients in the clinical setting, which might be answered through basic and translational science. Remarkably, curiosity is the cornerstone

for the development and maintenance of professional competence through lifelong learning and is indispensable to all physicians. For this reason, the ESC promotes and supports educational and training activities.

Preserve and protect research time

Medical scientists should have substantial protected time for research. However, the requirements of academic publications and tenure track records can conflict with patient care duties and responsibilities. Physicians involved in basic and clinical research often feel overwhelmed by clinical and research commitments, which contributes to stress and burnout, affecting not only their well-being but also the quality of patient care as well as their research.^{61,62} To thrive and achieve their full potential, medical scientists need carefully negotiated protected time and clinical coverage agreements, guaranteed access to career development opportunities, training, and educational resources, critical to research success.⁶³

Conclusions

While the outcomes of basic research may not be immediately apparent, they are crucial groundwork for future applications that can have long-lasting societal benefits.¹⁰ Engaging in discovery, translational, and clinical research during medical or cardiology education and training represents a unique opportunity and challenge for students and trainees to test their talents, passion, and curiosity, explore cardiology subspecialties, identify specific interests and align their research accordingly in subsequent years, ultimately contributing to the ideal development of a well-rounded and credentialed physician.^{64,65} We expect that the skills and knowledge gained through a combined multifaceted approach will reduce the gap between cardiovascular research and clinical care, translating into better patient care and resolution of complex clinical questions.

Acknowledgements

This manuscript is dedicated to the memory of I.A., our dear colleague and friend, whose sudden passing left a shocking void in our hearts and in the scientific community. This manuscript, which builds upon ideas and insights that Ioanna deeply cared about, is a testament to her vision and tireless efforts for translation of scientific concepts to young cardiologists. May her example continue to resonate as a legacy. We are indebted to the people who helped us to create [Table 1](#). More specifically, to Alexandru Scafa Udriste, from the Faculty of Medicine, University of Medicine and Pharmacy 'Carol Davila,' Bucharest, Romania; Ana Lebreiro, from the São João Hospital Centre, Porto, Portugal; Andreja Sinkovič, from the University Clinical Centre Maribor, Maribor, Slovenia; Beata Lewandowska, President of the Polish Cardiac Society, Poland; Edina Cenko, from the Department of Medical and Surgical Sciences, University of Bologna, Italy; Gro Egholm, from the Odense University Hospital, Denmark; Henrique Girao, from the University of Coimbra, Portugal; Joanna Hlebowicz, from the Lund University, Lund, Sweden; Maryanne Caruana from the Department of Cardiology at Mater Dei Hospital, Malta; Olivija Dobilienė, from the Lithuanian University of Health Sciences, Kaunas, Lithuania; Tomasz Pawłowski, Secretary of the Polish Cardiac Society, Poland; and President of Finnish Cardiac Society, Tuomas Kiviniemi, from the Turku University Hospital and University of Turku, Finland. We also thank the contribution of the following students from the

3rd International Summer School 'Teaming up for global cardiovascular risk reduction pathophysiology, pharmacology and therapeutics', Padova, Italy: Bob Clymans (Belgium), Izabella Tacinova (Czech Republic), Ksenia Pribõlouskaja (Estonia), Sabina Brus (Slovenia), and Wiktoria Sobocińska (Poland).

Supplementary data

Supplementary data are available at [European Heart Journal](#) online.

Declarations

Disclosure of Interest

Centro Cardiologico Monzino, IRCCS, receives funding from the pharmaceutical Companies Dompé and Chiesi Farmaceutici for MPE Group to perform research, respectively, on the angiogenic function of proprietary peptides and variants of the SARS-CoV-2 Spike protein variant, and on the effect of proprietary molecules on calcification of aortic valve interstitial cells. A.M. received consultant fees from BioMarin Pharmaceutical Inc. RAB reports research or educational grants received by the institution of employment from Abbott Vascular, Biosensors, Boston Scientific, and Translumina, which do not impact his personal remuneration and are unrelated to the work under consideration. K.S.-B. has no competing interest directly related to this work but has grants from Novartis and received speaker's honoraria from Novartis. T.K. is holding patents for the use of malonate against ischaemia/reperfusion injury and co-founder of Camoxis Pharmaceuticals. C.G.T. reports honoraria or consultation fees from VivaLyfe, Univers Formazione, Solaris, Summeet, AstraZeneca, Myocardial Solutions, and Medtronic; funding from Amgen and MSD; and is listed as an inventor of two patents related to HF, outside the submitted work. P.Vranckx. reports grants or contracts from Bristol Myers Squibb, Janssen, and Daiichi Sankyo. He reports honoraria from Oxford University Press (Editor in Chief *European Heart Journal Acute Cardiovascular Care*, *ESC Textbook on Acute Cardiovascular and Intensive Care Medicine*). Dr Rubini Gimenez reports speakers or advisors' honoraria from Beckman Coulter, Roche, Ortho Clinical Diagnostics, Quidel, Spinchip, and Siemens, outside the current work. The other authors have nothing to disclose.

Data Availability

No data were generated or analysed for this manuscript.

Funding

Instituto de Salud Carlos III (PI21/00637) to G.C.-B.; Agence Nationale de la Recherche (ANR- 20-CE14-0013-01) and Fondation de France (00107048) to MePa; Hellenic Foundation for Research and Innovation (H.F.R.I.) under the '2nd Call for H.F.R.I. Research Projects to support Faculty Members & Researchers' (project number: 03586) to I.A.; Swiss National Science Foundation (320030_215035), Fondation Pierre Mercier Pour la Science and Swiss Heart Foundation (FF19110) to A.M.; Swiss Heart Foundation and Swiss National Foundation (P400PM_180828) to M.R.; Italian Ministry of Health (Ricerca Corrente, ricerca 5 per mille, PNRR programme) to MaPe; Ministero dell'Università e della Ricerca grant(2020YRETTX_003), and Next Generation EU grant 'National Center for Gene Therapy and Drugs based on RNA Technology' (CN00000041, concession number 1035 of 17 June 2022-PNRR MUR—

M4C2—Investment 1.4 Call 'National Centers'), project code (CUP) E63C22000940007 to C.P.; German Research Foundation (DFG—Deutsche Forschungsgemeinschaft) with the Collaborative Research Center (CRC) 1470 (SFB/CRC 1470/A07 and project 536819681), and the Deutsche Krebshilfe (project 70115119) to S.V.L.; DFG with the Collaborative Research Center (CRC) 141213 (SFB/CRC 1213/B07), the RTG 2824/P04 and the project number 471241922 to K.S.-B.; The Ministry for Innovation and Technology in Hungary provided funding to this study under the 2020-1.1.5-GYORSÍTÓSAV call programme (2020-1.1.5-GYORSÍTÓSAV-2021-00011), the TKP2021-EGA funding scheme (TKP2021-EGA-23), and the Thematic Excellence Programme (TKP/ITM/NKFIH) to Z.G. This study was also supported by project no. RRF-2.3.1-21-2022-00003 'National Heart Laboratory, Hungary' implemented with the support provided by the European Union, by the National Research, Development and Innovation Office of Hungary (NKFI; VEKOP-2.3.2-16-2016-00002 and VEKOP-2.3.3-15-2017-00016), as well as by the Hungarian National Scientific Research Fund (K_21-139105) to Z.G.; Dutch Heart Foundation (2020-2020B003, DnAFix) and NWA-ORC project CIRCULAR NWO (NWA.1389.20.157) to B.B. DZHK (German Centre for Cardiovascular Research—81X3100210; 81X2100282) to M.T. C.G.T. is supported by two grants from the Italian Ministry of Health (PNRR-MAD-2022-12376632 and RF-2016-02362988). CIBEROBN is an initiative of the Instituto de Salud Carlos III (ISCIII), Spain, by the European Regional Development Fund (ERDF) 'A way of making Europe'.

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